```
clc; clear; close all;
```

Butterworth Filter Calculations

```
% 1) Transfer Function (From Exam 3 Cheat-Sheets)

% H(s) = Kwc^3 // (s+wc)(s^2 + s + wc^2)
% for wc = 2*pi*2000
%
%      = 100*wc^3 // (s^3 + 2*wc*s^2 + (2*wc^2)s + wc^3)


% 2) Plots

% Had to ask for help programming the symbolic section & group delay of this
part:
% "How do I use an algebraic transfer function's frequency response in
MATLAB?"
% "How do I plot group delay in MATLAB & show it's corner frequency's
response?"

s = tf('s'); wc = 2*pi*2000; H = 100*wc^3/(s^3 +
2*wc*s^2 + 2*wc^2*s + wc^3);

f = logspace(1,5,600);
w = 2*pi*f;

[mag, phase] = bode(H,w);
mag = squeeze(mag); phase
= squeeze(phase); %
Magnitude

figure(1); subplot(4,1,1) semilogx(f,20*log10(mag)); hold
on; plot(2000,37,'ro'); grid on; title('3rd-Order
Butterworth Low-pass Filter Magnitude');
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');

% Phase

subplot(4,1,2)
semilogx(f, phase);
hold on; plot(2000, -
135, 'ro') grid on;
title('3rd-Order Butterworth Low-pass Filter Phase');
```

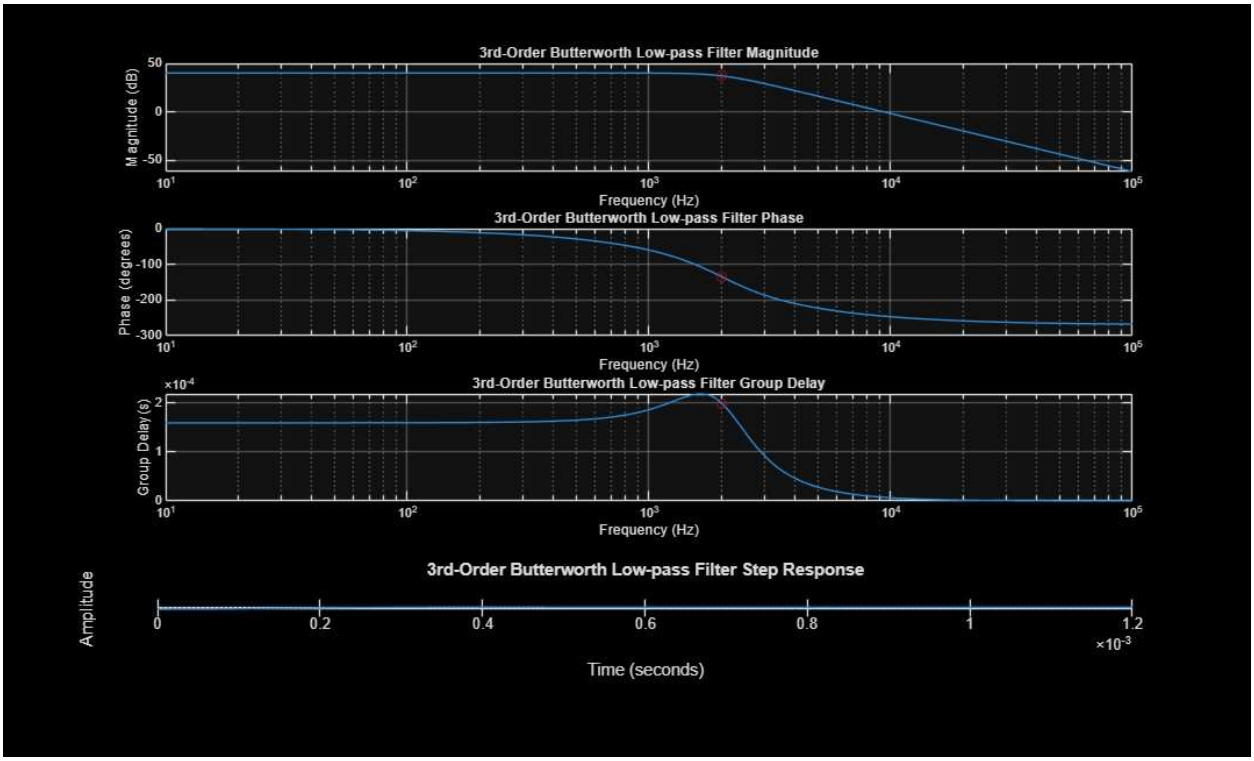


```
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');

% Group Delay
subplot(4,1,3)
```

```
gd = -gradient(unwrap(phase*pi/180),w); semilogx(f,gd);
hold on; plot(2000,interp1(f,gd,2000),'ro'); grid on;
title('3rd-Order Butterworth Low-pass Filter Group Delay');
xlabel('Frequency (Hz)'); ylabel('Group Delay(s)');

subplot(4,1,4); step(H); grid on; title('3rd-Order
Butterworth Low-pass Filter Step Response'); xlabel('Time');
ylabel('Amplitude');
```



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1)

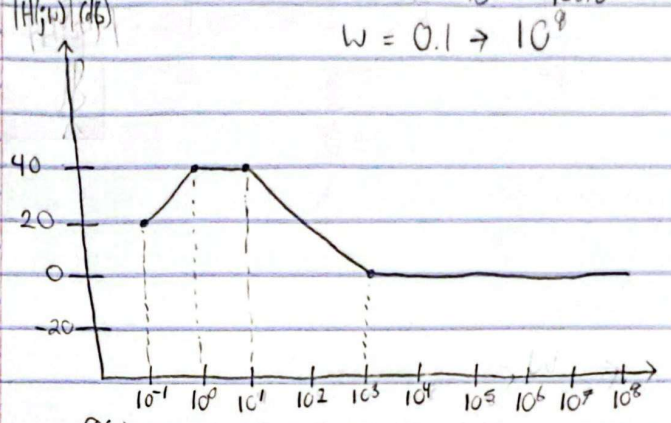
$$H(s) = \frac{100s\left(1+\frac{s}{1000}\right)}{(s+1)\left(\frac{s}{10}+1\right)}$$

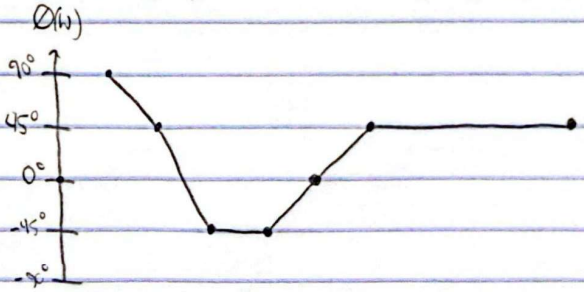
Zeroes @ 0, 1000

→ Poles @ 1, 10

$K = 100 = 40dB$

$W = 0.1 \rightarrow 10^9$

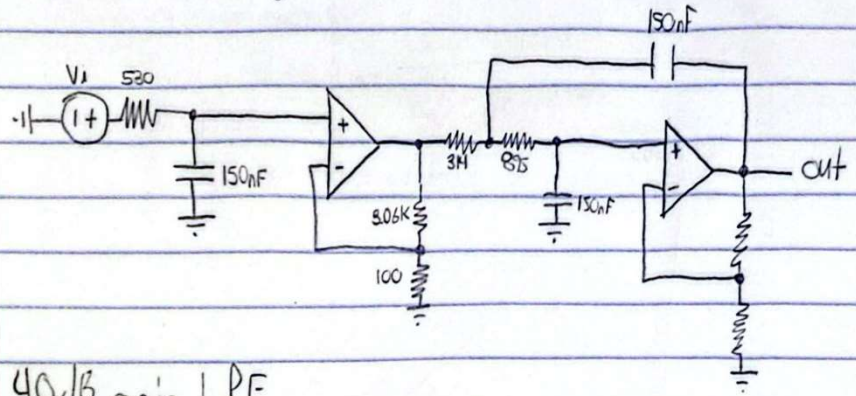




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Analog Filter Wizard By AD

2)

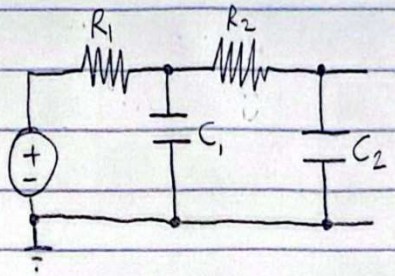


40dB gain LPF

$f_c = 2000Hz$

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3)



Previously derived: HW5 Q6, to show loading effect w/o op-amp.

a) $$H(s) = \frac{1/R_1R_2C_1C_2}{s^2 + s\left(\frac{1}{R_1C_1} + \frac{1}{R_2C_2} + \frac{1}{R_2C_1}\right) + 1/R_1R_2C_1C_2}$$

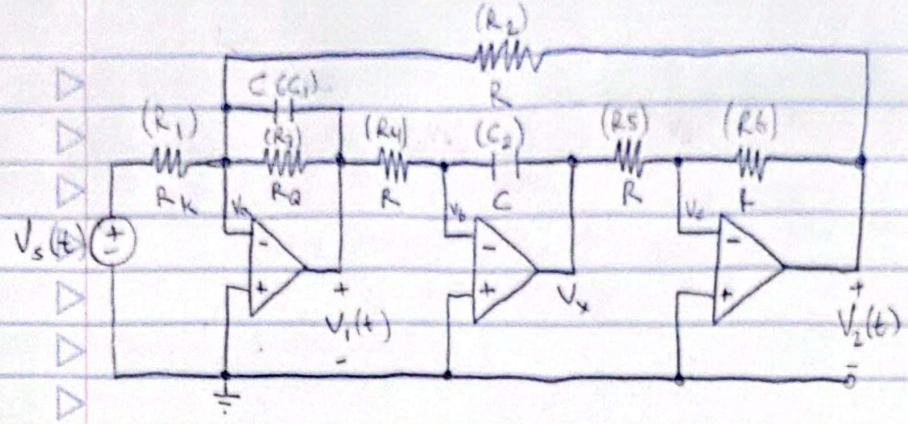
b) $$H(s) = \frac{1/R \cdot RK \cdot C \cdot \frac{C}{K}}{s^2 + s\left(\frac{1}{RC} + \frac{1}{RK(C/K)} + \frac{1}{RKC}\right) + \frac{1}{RRCC}}$$

$$= \frac{1/R^2C^2}{s^2 + s\left(\frac{2}{RC} + \frac{1}{RCK}\right) + \frac{1}{R^2C^2}}$$, becomes ideal as $k \to \infty$

$|H(\omega)| =$

k	$2+\frac{1}{k}$
0.1	12
1	3
10	2.1
100	2.01
∞	2 ← ideal

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EQ1: $\frac{V_a - V_i}{R_k} + \frac{V_a - V_{O2}}{R} + \frac{V_a - V_{O1}}{R_Q} + (V_a - V_{O1})sC = 0$

EQ2: $\frac{V_b - V_{O1}}{R} + (V_b - V_x)sC = 0$

EQ3: $\frac{V_c - V_x}{R} + \frac{V_c - V_{O2}}{R} = 0 \qquad V_a, V_b, V_c = 0$

EQ1: $\frac{V_i}{R_k} + \frac{V_{O2}}{R} + \frac{V_{O1}}{R_Q} + V_{O1}sC = 0$

EQ2: $\frac{V_{O1}}{R} + V_x sC = 0 \rightarrow V_x = -\frac{V_{O1}}{sRC} \Rightarrow V_{O2} = \frac{V_{O1}}{sRC}$

EQ3: $\frac{V_{O2}}{R} + \frac{V_x}{R} \rightarrow V_{O2} = -V_x$

$EQ_1: \frac{V_i}{R_k} - \frac{V_{O1}}{sRC} \cdot \frac{1}{R} + \frac{V_{O1}}{R_Q} + V_{O1}sC = 0$

$\frac{V_i}{R_k} = -V_{O1}\left(\frac{1}{sR^2C} + \frac{1}{R_Q} + sC\right) \rightarrow H(s) = -\frac{1}{R_k}\left(\frac{1}{sR^2C} + \frac{1}{R_Q} + sC\right) \cdot \frac{R^2C}{sR_k}$ (R²C / sRk)

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$$\frac{V_{o1}}{V_i} = \frac{-sR^2C/R_k}{s^2R^2C^2 + sR^2C/R_Q + 1} = -\frac{s(1/R_kC)}{s^2 + s\left(\frac{1}{CR_Q}\right) + \frac{1}{R^2C^2}}$$

Bandpass

$$\frac{V_{o2}}{V_i} = \frac{V_{o1}}{V_i} \cdot \frac{1}{sRC} = -\frac{1/R_kRC^2}{s^2 + s(1/R_QC) + \frac{1}{R^2C^2}}$$ LPF

$$W_0 = 2\pi(1250) = 2500\pi \approx 7853 = \frac{1}{\sqrt{R_2R_4C_1C_2}}$$

$\frac{R_2}{R_1} = 5$, $R_2 = 10k$, $R_1 = 2k$

$$Q = \sqrt{\frac{R_3^2C_1}{R_2R_4C_2}} = \frac{1}{\sqrt{2}} \rightarrow C_1 = C_2 = 10nF \rightarrow R_2 = R_4 = 10k$$

$$Q = \sqrt{\frac{R_3^2}{R_2^2}} = R_3/R_2 \rightarrow R_3 = 7.071k$$

From previous:

R_1	2k
R_2	10K
R_3	7.071k
R_4	10k
C_1	10 nF
C_2	10nF

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